

Joohyun Lee

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RESEARCH INTEREST

Theoretical & Computational Astrophysics

cosmological simulation of the Epoch of Reionization by numerical radiation hydrodynamics;
testing dark matter models with cosmological structure formation;
galaxy formation & evolution; machine learning applied to simulation data analysis

EDUCATION

Ph.D. in Astronomy , The University of Texas at Austin (Supervisor: Prof. Paul Shapiro)	2021 – present
M.A. in Astronomy , The University of Texas at Austin	2021 – 2023
B.Sc. in Physics & Electrical and Computer , Seoul National University	2014 – 2021 (incl. military service)

PROFESSIONAL APPOINTMENTS

Graduate Student Fellow , UT Austin	09/2021 – 08/2022, 06/2026 – present
Graduate Teaching Assistant , UT Austin	09/2025 – 05/2026
NASA FINESST Fellow , UT Austin (w. Prof. Paul Shapiro)	09/2022 – 08/2025
Undergraduate Research Associate , SNU (w. Prof. Jong-Hak Woo)	09/2020 – 02/2021
Undergraduate Research Associate , SNU (w. Prof. Ji-hoon Kim)	09/2019 – 08/2020, 03/2021 – 08/2021

AWARDS

Visualizing Science 2025 Editor's choice , The University of Texas at Austin	03/2026
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AWARDED FELLOWSHIPS & SCHOLARSHIPS

Outstanding Research Fellowship (\$53k) , The University of Texas at Austin	09/2026 – 08/2027
Craig Wheeler Fellowship (Summer Fellowship), The University of Texas at Austin	05/2026 – 08/2026
FINESST Fellowship (\$150k) , NASA	09/2022 – 08/2025
Dean's Excellence Fellowship , The University of Texas at Austin	09/2021 – 08/2022
Presidential Science Scholarship (~ \$40k) , Korea Student Aid Foundation	03/2014 – 08/2020

AWARDED COMPUTING TIME

Stampede3 (~ 50k CPU hours) , PI, NSF ACCESS Discover	06/2025 - 06/2026
Stampede3 (~ 6k CPU hours) , PI, NSF ACCESS Explore	03/2024 - 06/2025

TEACHING EXPERIENCE

AST 301 Introduction to Astronomy , Graduate Teaching Assistant	Spring 2026
AST 376C Cosmology , Graduate Teaching Assistant	Fall 2025
AST 382C Astrophysical Gas Dynamics (Graduate Course), Grader	Fall 2024

PUBLICATIONS (ADS Library) (Google Scholar)

5. Lee, E. H., **Lee, J.**, Shapiro, P. R., et al., “Line Intensity Mapping Prediction from the Cosmic Dawn III Simulation for $H\alpha$ from Galaxies and the Intergalactic Medium During the Epoch of Reionization”, *RNAAS* 9 (2025) 96, *astro-ph:2504.15821*
4. Ocvirk, P., Lewis, J. S. W., Conaboy, L., ..., **Lee, J.**, et al., “Dust-UV offsets in high-redshift galaxies in the Cosmic Dawn III simulation”, *A&A* 703 (2025) A98, *astro-ph:2409.05946*
3. **Lee, J.**, Shin, E. -j., Kim, J. -h., Shapiro, P. R., & Chung, E., “Multiple Beads on a String: Dark-matter-deficient Galaxy Formation in a Mini-Bullet Satellite–Satellite Galaxy Collision”, *ApJ* 966 (2024) 72, *astro-ph:2312.11350*
2. **Lee, J.**, Shin, E. -j., & Kim, J. -h., “Dark Matter Deficient Galaxies And Their Member Star Clusters Form Simultaneously During High-velocity Galaxy Collisions In 1.25 pc Resolution Simulations”, *ApJL* 917 (2021) L15, *astro-ph:2108.01102*
1. Shin, E. -j., Jung, M., Kwon, G., Kim, J. -h., **Lee, J.**, Jo, Y., & Oh, B. K., “Dark Matter Deficient Galaxies Produced Via High-velocity Galaxy Collisions In High-resolution Numerical Simulations”, *ApJ* 899 (2020) 25, *astro-ph:2007.09889*

RESEARCH EXPERIENCE

Graduate Student, The University of Texas at Austin (w. Prof. Paul Shapiro)

09/2021 –

- Multiple Beads-on-a-string: Dark-Matter-Deficient Galaxies Produced in a Mini-Bullet Galaxy-galaxy Collision
 - Ran a suite of N -body/hydro simulations of galaxy collisions of satellites of a massive host galaxy
 - Performed orbit integration of resulting dark-matter-deficient galaxies to compare with recent observations of trail of galaxies
 - Demonstrated simultaneous formation of multiple aligned dark-matter-deficient galaxies that match observation
- The Inhomogeneous Rise of Metallicity and Reionization During the Epoch of Reionization
 - Studied the Cosmic Dawn (CoDa) III simulation to quantify the inhomogeneity of the metal enrichment process in the intergalactic medium/circumgalactic medium during the EoR
 - Studied CoDa III simulation to measure reionization H II bubble growth
- Modeling UV luminosity function in a large box cosmological simulation
 - Studied halo UV luminosity statistics in the CoDa III simulation by varying subgrid dust and stellar physics parameters
 - Applied friends-of-friends (FoF) halo finding algorithm to only star particles to mock observational approach of identifying galaxies in the early Universe
- Study of hydrogen Balmer line intensity mapping and statistical analysis
 - Modeled hydrogen Balmer ($H\alpha$, $H\beta$) line emission in the CoDa III simulation and statistics targeted for future space missions

Research Associate, Computational Cosmology Group, SNU (w. Prof. Ji-hoon Kim)

09/2019 – 08/2021

- Estimating Galactic Baryonic Properties from Their Dark Matter Using Machine Learning
 - Applied trained machine to the cosmological simulation halo catalog (IllustrisTNG simulation)
 - Computed and compared two-point correlation function in IllustrisTNG halo catalog and machine-predicted halo catalog
- Dark-Matter-Deficient Galaxies Produced via High-velocity Galaxy Collision in Cosmological Simulation
 - Studied IllustrisTNG catalog to find high-speed collision events of dwarf galaxies to compare with idealized simulation
- pc-scale Simulation of Simultaneous Formation of Dark-Matter-Deficient Galaxies and Star Clusters
 - Ran a suite of 1.25 pc-resolution galaxy collision simulations with different merger configurations and feedback schemes
 - Resolved and tracked the formation process of dark-matter-deficient galaxies and massive star clusters

Research Associate, AGN Research Group, SNU (w. Prof. Jong-Hak Woo)

09/2020 – 02/2021

- Calibrated and Applied Novel Method of Measuring Star Formation Rates in Active Galactic Nuclei
 - Tested Oxygen emission line flux as SFR indicator by statistically analyzing SDSS spectroscopy data and IR surveys
 - Investigated correlation between gas outflow strength from AGNs and star formation of host galaxies

INVITED TALKS & PRESENTATIONS (talk slides linked)

- Theoretical Astroparticle and Cosmology Symposium in Texas (TACOS) 2025 10/2025
Results from CoDa III: Inhomogeneous Reionization and Chemical Enrichment with Observational Signatures
- 10 Years of Ultra-Diffuse Galaxies: Past Perspectives and Future Prospects 08/2025
Multiple Beads-on-a-string: Dark-matter-deficient Galaxy Formation in a Mini-Bullet Satellite-satellite Galaxy Collision

CONTRIBUTED TALK & POSTER PRESENTATIONS

- UT Austin Extragalactic/Cosmology Seminar 05/2026
Galaxy Properties in the Early Universe Studied with CoDa III Simulation
- CFC2025 Inaugural Conference 05/2025
The Inhomogeneous Reionization Rise of Metallicity During the Epoch of Reionization
- APS Global Physics Summit 2025 03/2025
Multiple Beads-on-a-string: Dark-matter-deficient Galaxy Formation in a Mini-Bullet Satellite-satellite Galaxy Collision
- UT Austin Extragalactic/Cosmology Seminar 11/2024
The Inhomogeneous Reionization Rise of Metallicity During the Epoch of Reionization + Other Results from CoDa III
- CCA-CFC Joint Workshop 11/2024
The Inhomogeneous Rise of Metallicity During the Epoch of Reionization in the CoDa III Simulation
- Cosmic Dawn at High Latitudes Conference 06/2024
The Inhomogeneous Rise of Metallicity During the Epoch of Reionization in the CoDa III Simulation
- First Stars VII Conference (poster & lightning talk) 05/2024
The Inhomogeneous Rise of Metallicity During the Epoch of Reionization in a Large-scale EoR Simulation
Cosmic Dawn ("CoDa") III
- Ramses User Meeting 2024 04/2024
Multiple Beads-on-a-string: Dark-matter-deficient Galaxy Formation in a Mini-Bullet Satellite-satellite Galaxy Collision
- UT Austin Extragalactic/Cosmology Seminar 11/2023
The Inhomogeneous Rise of Metallicity During the Epoch of Reionization
- FirstLight Conference (poster & lightning talk) 06/2023
The Inhomogeneous Rise of Metallicity During the Epoch of Reionization
- UT Austin Extragalactic/Cosmology Seminar 05/2023
Understanding Galaxy and Structure Formation in the Cold Dark Matter Universe
- IAUGA 2022 (poster) 08/2022
Dark Matter Deficient Galaxies (DMDGs) And Their Member Star Clusters (SCs) Form Simultaneously during High-velocity Galaxy Collisions
- APS April Meeting 2022 04/2022
Dark Matter Deficient Galaxies (DMDGs) And Their Member Star Clusters (SCs) Form Simultaneously during High-velocity Galaxy Collisions
- Galaxy Evolution Workshop 2021, ASIAA 02/2022
Dark Matter Deficient Galaxies (DMDGs) And Their Member Star Clusters (SCs) Form Simultaneously during High-velocity Galaxy Collisions in 1.25 pc Resolution Simulations
- Numerical Galaxy Formation Mini-Workshop, SNU 01/2022
Dark Matter Deficient Galaxies (DMDGs) And Their Member Star Clusters (SCs) Form Simultaneously during High-velocity Galaxy Collisions in 1.25 pc Resolution Simulations
- SAZERAC-SIPS Early Galaxy Formation Near and Far — Preparing for a Long Journey with JWST 12/2021
Dark Matter Deficient Galaxies (DMDGs) And Their Member Star Clusters (SCs) Form Simultaneously during High-velocity Galaxy Collisions in 1.25 pc Resolution Simulations
- The 1st KIAA Forum on Gas in Galaxies for Early Career Scientists (KooGiG-Junior workshop) 10/2021
Dark Matter Deficient Galaxies (DMDGs) And Their Member Star Clusters (SCs) Form Simultaneously during High-velocity Galaxy Collisions in 1.25 pc Resolution Simulations

CONTRIBUTED TALK & POSTER PRESENTATIONS (CONTINUED)

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| • UT Austin Extragalactic/Cosmology Seminar
<i>Dark Matter Deficient Galaxies (DMDGs) And Their Member Star Clusters (SCs) Form Simultaneously during High-velocity Galaxy Collisions in 1.25 pc Resolution Simulations</i> | 09/2021 |
| • AGORA WORKSHOP 2021
<i>Dark Matter Deficient Galaxies (DMDGs) And Their Member Star Clusters (SCs) Form Simultaneously during High-velocity Galaxy Collisions in 1.25 pc Resolution Simulations</i> | 08/2021 |

ACADEMIC & DEPARTMENTAL SERVICE

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| Hosting Galaxy Formation & Cosmological Simulation Seminar in UT Astro. Dept. | 06/2025 – |
| Co-captain of UT Astro. Dept. Softball Team | 03/2024 – |

ADVISING & RESEARCH MENTORING

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| Rishita Agarwal (undergrad at UT), as a research mentor | 01/2026 – |
| Eugene Lee (undergrad at UT), as a research mentor, co-authored a research note | 05/2024 – |
| Jimin Shin (undergrad at UT, Yonsei), as a research mentor | 09/2025 – 12/2025 |
| Jany Esquivel (NSF REU undergrad at UT), as an informal research mentor | 06/2024 – 07/2024 |
| Eunwoo Chung (undergrad at SNU), as a research mentor, co-authored a paper | 01/2023 – 12/2023 |

COMPUTING SKILLS & EXPERIENCES

- Languages:** Python, C, C++, LaTeX (skilled); Fortran, MATLAB, Mathematica, HTML, Markdown (familiar); IDL, RISC-V assembly language (basic)
- Astrophysical Simulation Codes:** Enzo, Ramses, Ramses-CUDATON, Gadget2, GIZMO
- Astrophysical Simulation Initial Condition Generators:** MUSIC, DICE
- Simulation Analysis Codes:** yt, Tools21cm,
- Machine Learning Frameworks:** JAX, PyTorch (familiar); TensorFlow (basic)
- High Performance Computing Experience:**
- Frontera (CentOS), Stampede3 (Rocky), Stampede2 (Red Hat), Vista (Rocky), Texas Advanced Computing Center
 - Andes, Oak Ridge National Laboratory (Fedora)
 - Local cluster of Computational Cosmology Group, Seoul National University (CentOS)
 - Nurion, Korea Institute of Science and Technology Information (CentOS)

MENTORING & OUTREACH

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| GUMMY Mentor in UT Astro. Dept. | 08/2023 – 12/2025 |
| Informal Mentor for Summer NSF REU Scholars | 2022, 2023, & 2024 |
| Korea Student Aid Foundation Science Teaching Service Organization | 01 – 02/2015 |
| Habitat for Humanity in Cebu, Philippines | 02/2016 |
| Military Service at Korean Air Force 5th Air Mobility Wing | 05/2017 – 04/2019 |

OTHER SKILLS

- Languages: Korean (native), English, Japanese (fluent)